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SQL Aggregate Functions & operations

1. SELECT DISTINCT department
FROM students,

department
IT
HR
Finance

2. SELECT department, AVG(age) AS avg-age
FROM students
GROUP BY department;

department	AVG-age
IT	20,5
HR	22
Finance	23

3. SELECT department, Count(*) AS student-count
FROM student
GROUP BY department
HAVING count(*) > 1;

department	student-count
IT	2
HR	2

4. SELECT student_id,
 name,
 age,
 department
 FROM Students
 WHERE age BETWEEN 21 AND 23;

student_id	name	age	department
2	Bob	22	HR
3	Charlie	21	IT
4	Diana	23	Finance
5	Eve	22	HR

5. SELECT student_id,
 name,
 age,
 department
 FROM Students
 WHERE department IN ('IT', 'HR') AND age > 21

Student-ID	Name	age	department
2	Bob	22	HR
5	EVE	22	HR

6. SELECT department, SUM(Credits) AS total_Credits
 FROM Courses
 GROUP BY department
 HAVING SUM(Credits) > 5;

department	total_credits
IT	11

7. SELECT course_id,
course_name,
department,
credits

FROM Courses

WHERE credits < 4;

Course ID	Course name	department	Credits
101	SQL Basics	IT	3
104	Excel	Finance	2
105	Statistics	HR	3

8. SELECT course_id,
course_name,
credits

FROM Courses

ORDER BY credits DESC

LIMIT 3;

Course id	Course-name	Credits
102	Python	4
103	Data Science	4
101	SQL Basics	3

department	total_credits
IT	11

7. SELECT course_id,
course_name,
department,
credits

FROM Courses

WHERE credits < > 4;

Course ID	Course name	department	Credits
101	SQL Basics	IT	3
104	Excel	Finance	2
105	Statistics	HR	3

8. SELECT course_id,
course_name,
credits

FROM Courses

ORDER BY credits DESC

LIMIT 3;

Course id	Course-name	Credits
102	Python	4
103	Data Science	4
101	SQL Basics	3

9. SELECT MAX(grade) AS max-grade,
MIN(grade) AS min-grade,
AVG(grade) AS AV-grade
FROM Enrollments

Max-grade	min-grade	AV-grade
90	78	84,6

10. SELECT course_id, COUNT(*) AS enrollment-count
FROM enrollments
GROUP BY course_id;

Course_id	enrollment-count
101	1
102	1
103	1
104	1
105	1

11 SELECT department, SUM(salary) AS total-salary,
SUM(bonus) AS total-bonus
FROM salaries
GROUP BY department;

department	total-salary	total-bonus
IT	122000	105000
HR	109000	7500
Finance	70000	6000

12 SELECT department, AVG (salary) AS avg - salary
 FROM Salaries
 GROUP BY department
 HAVING AVG (salary) > 55000;

department	avg - salary
IT	61000
Finance	70000

13 SELECT employees Id, name,
 salary,
 bonus
 (Salary + bonus) AS total - compensation
 FROM Salaries
 WHERE (Salary + bonus) > 60000

employee Id	name	salary	bonus	total compensation
1	Tom	60000	5000	65000
3	Spalke	70000	6000	76000
4	Tyke	62000	8500	70500

14 SELECT department, sum (budget) AS total - budget,
 avg (budget) AS avg - budget
 FROM projects
 GROUP BY department
 HAVING avg (budget) > 70000;

department	total-budget	Avg-budget
IT	270000	135000
Finance	80000	80000

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15 SELECT project-id,
           project-name,
           department,
           budget
FROM projects
WHERE budget BETWEEN 50000 AND 120000
AND department <> 'marketing';

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Project-id	Project-name	department	Budget
1	AI app	IT	120000
2	Payroll System	Finance	80000
3	HR Portal	HR	50000